Minimal Geth Fork as EVM Execution Engine for Enterprise zkEVM L2: Architecture Analysis, Opcode Constraint Mapping, and Trace Generation

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***Abstract*—Enterprise zkEVM Layer 2 architectures require an EVM execution engine capable of processing transactions with full opcode compatibility while simultaneously producing structured execution traces for downstream zero-knowledge proof generation. We investigate the feasibility of using a minimal fork of go-ethereum (Geth) as the execution engine for Basis Network, an enterprise zkEVM L2 deployed on Avalanche. Through systematic analysis of five production systems—Polygon CDK (cdk-erigon), Scroll (scroll-geth), zkSync Era (EraVM), Optimism (op-geth), and Reth—we establish that no production zkEVM uses the EVM directly for proving; all separate execution from proof generation. We map 75 Cancun EVM opcodes to their zero-knowledge constraint costs across both R1CS and PLONKish arithmetizations, identifying KECCAK256 ($\sim$150,000 R1CS constraints) and SLOAD/SSTORE (255 Poseidon operations each) as the dominant proving costs. We design a custom ZK tracer leveraging Geth's `core/tracing` hooks API that captures only state-modifying operations (storage, balance, nonce, logs), achieving a projected 10–30% execution overhead compared to 10–50$\times$ for full opcode tracing. Projected benchmarks based on published Geth throughput data indicate 5,000–15,000 simple transfers per second and 2,000–8,000 storage writes per second without tracing, with selective ZK tracing reducing throughput by 10–25% and 15–30% respectively—well above the 1,000 tx/s hypothesis threshold. The key architectural recommendation is to import Geth as a Go module dependency (Strategy A) rather than maintaining a full repository fork (Strategy B), leveraging the `vm.StateDB` interface to integrate a custom Poseidon-based sparse Merkle tree without modifying Geth internals. This approach is validated by Optimism's op-geth, which achieves a complete L2 execution layer with only 34 lines of EVM modifications.**

***Index Terms*—zkEVM, EVM execution, go-ethereum, execution traces, ZK witness generation, opcode constraints, enterprise blockchain, Avalanche L2**

I. Introduction

The construction of a zero-knowledge Ethereum Virtual Machine (zkEVM) Layer 2 requires solving a fundamental architectural problem: how to execute EVM bytecode with full compatibility while simultaneously producing the structured execution traces that a zero-knowledge prover requires for witness generation. This problem sits at the intersection of systems engineering (EVM execution performance, state management, tracing infrastructure) and applied cryptography (constraint system design, hash function selection, proof generation).

Basis Network is an enterprise blockchain platform deployed on Avalanche, targeting Latin American industrial and commercial clients. The platform operates an application-specific L1 (Subnet-EVM, Chain ID 43199) with zero-fee permissioned transactions. The next phase of development—an enterprise zkEVM L2—requires each client enterprise to operate its own execution chain with ZK validity proofs verified on the Basis Network L1. This architecture demands an execution engine that satisfies three properties simultaneously:

1) **Full Cancun opcode compatibility**: Enterprise smart contracts compiled with Solidity 0.8.24+ targeting the Cancun EVM version must execute identically to mainnet Ethereum, including transient storage (EIP-1153), memory copy (EIP-5656), and blob operations (EIP-4844).
2) **Structured trace generation**: Every state-modifying operation (storage reads/writes, balance changes, nonce updates, log emissions, contract creation) must be captured in a format consumable by the downstream Rust-based ZK prover via gRPC.
3) **Enterprise-grade throughput**: The execution engine must sustain at least 1,000 transactions per second with trace generation enabled, matching the throughput requirements of industrial traceability and commercial ERP workloads.

The research question addressed by this paper is:

Can a minimal fork of go-ethereum execute EVM transactions with custom state management, producing execution traces necessary for ZK witness generation, while maintaining 100% Cancun opcode compatibility and processing 1,000+ simple transactions per second?

This question is investigated through a systematic analysis of five production L2 architectures, a complete mapping of Cancun EVM opcodes to their zero-knowledge constraint

costs, and the design of a custom ZK tracer that leverages Geth's event-driven tracing hooks API.

The remainder of this paper is organized as follows. Section II surveys related work on zkEVM architectures and EVM fork strategies. Section III presents the system model and architectural constraints. Section IV describes the experimental methodology, including the opcode constraint mapping and tracer design. Section V details the experimental setup. Section VI presents results including projected throughput benchmarks and constraint cost analysis. Section VII discusses implications, invariants, and limitations. Section VIII concludes with recommendations for the Basis Network execution engine.

II. Related Work

A. zkEVM Type Spectrum

Buterin [1] introduced the zkEVM type classification ranging from Type 1 (fully Ethereum-equivalent) to Type 4 (language-compatible only). Chaliasos et al. [2] formalized this spectrum and benchmarked production zkEVM systems, finding that Type 2–3 systems (Polygon zkEVM, Scroll) achieve practical proving times of 190–200 seconds per batch while maintaining high EVM compatibility. The constraint-level analysis by Burchardt et al. [3] demonstrates that PLONKish arithmetizations with custom gates reduce constraint counts by $5\times$–$30\times$ compared to R1CS for arithmetic-heavy opcodes.

B. Production zkEVM Architectures

Five production systems inform the architectural decisions in this paper.

Polygon CDK (cdk-erigon). Polygon migrated from a custom Go implementation (zkevm-node, archived February 2025) to an Erigon fork [4]. The execution layer uses Go (Erigon), while the proving layer is a separate C++ executor that processes transactions through zkASM (a custom assembly language). This separation achieves $10\times$ less disk usage and $150\times$ faster synchronization compared to the original zkevm-node. The key insight is that *the proving executor is architecturally separate from the EVM executor*.

Scroll (scroll-geth). Scroll maintains a direct fork of go-ethereum [5] that replaces the Merkle Patricia Trie (MPT) with a ZK-friendly Poseidon-based trie (zktrie), activated via `config.scroll.useZktrie`. However, the 2025 Euclid upgrade [6] migrates *back* to standard MPT combined with OpenVM (a general-purpose zkVM), suggesting that tightly coupling the state trie to the EVM fork creates unsustainable maintenance burden.

zkSync Era (EraVM). Rather than forking Geth, zkSync built a custom register-based virtual machine (EraVM) [7] optimized for ZK circuits. An EVM bytecode interpreter was added on top for compatibility. This approach trades compatibility (Type 4) for proving efficiency, with the Boojum 2.0 upgrade targeting native EVM execution.

Optimism (op-geth). Optimism's Geth fork [8] demonstrates the minimal viable diff: 16,881 lines added and 664 deleted against upstream Geth, with only 34 lines touching the EVM itself (precompile overrides, caller overrides). While op-geth is an optimistic rollup (no ZK proving), it establishes an upper bound on fork minimality.

Reth. Paradigm's Rust EVM implementation [9] achieves 100–200 MGas/s in live sync and up to 1.5 GGas/s with parallel execution (Gravity Reth) [10]. The revmc JIT compiler [11] further improves interpreted EVM performance by $1.85\times$–$19\times$. These benchmarks establish the performance ceiling for EVM execution engines.

C. EVM Tracing Infrastructure

Geth provides three tracing approaches with different performance characteristics. The `structLogger` captures every opcode with full stack, memory, and storage state, incurring 10–50$\times$ overhead [12]. The `callTracer` operates at call-frame granularity with lower overhead. Since Geth 1.14+, the `core/tracing` hooks API [13] provides event-driven callbacks (`OnOpcode`, `OnStorageChange`, `OnBalanceChange`, `OnNonceChange`) that enable selective capture of only the operations relevant to ZK witness generation. Production zkEVMs universally use custom compiled tracers rather than the built-in JavaScript-based tracing infrastructure.

D. ZK Constraint Costs

The Polygon zkEVM ROM specification [14] documents per-opcode ZK counter costs across five resource dimensions: arithmetic (`cnt_arith`), binary (`cnt_binary`), Keccak-f permutations (`cnt_keccak_f`), Poseidon operations (`cnt_poseidon_g`), and memory alignment (`cnt_mem_align`). KECCAK256 dominates with 192 arithmetic counters, 193 binary counters, and 2 Keccak-f permutations, translating to approximately 150,000 R1CS constraints per invocation [3]. Storage operations (SLOAD, SSTORE) require 255 Poseidon operations each for state trie traversal.

III. System Model

A. Target Architecture

The Basis Network enterprise zkEVM L2 operates a per-enterprise chain architecture where each client enterprise runs an independent L2 instance. Each instance consists of:

- A **Go-based sequencer/executor node** that receives transactions, executes them against a Poseidon-based sparse Merkle tree (SMT), and produces structured execution traces.
- A **Rust-based ZK prover** that consumes execution traces via gRPC, generates PLONK validity proofs, and returns proof artifacts.
- A **proof submitter** that posts validity proofs and state roots to the Basis Network L1 (Avalanche Subnet-EVM, Chain ID 43199).

This architecture separates execution (Go) from proving (Rust), matching the established pattern observed in all five production zkEVM systems surveyed in Section II.

B. EVM Execution Requirements

The execution engine must satisfy the following formal requirements:

1) **Opcode completeness**: For every Cancun EVM opcode $o \in O_{cancun}$ and valid input state σ, the execution result must be identical to the reference Geth implementation: $\forall o \in O_{cancun}, \sigma : \text{exec}_{basis}(o, \sigma) = \text{exec}_{geth}(o, \sigma)$.
2) **Trace completeness**: For every executed transaction tx, the trace $T(tx)$ must contain all state-modifying operations: $T(tx) \supseteq \{(op, addr, slot, val) \mid op \in \{\text{SLOAD, SSTORE, BALANCE_CHANGE, NONCE_CHANGE, LOG, CREATE}\}\}$.
3) **Throughput bound**: With selective ZK tracing enabled, throughput must exceed 1,000 transactions per second for simple transfers (21,000 gas each).

C. Geth Module Dependency Analysis

We analyze go-ethereum's module structure to identify the minimal set of packages required for standalone EVM execution with tracing.

TABLE I
GO-ETHEREUM MODULES REQUIRED FOR STANDALONE EXECUTION

Module	Purpose	LOC
`core/vm/`	EVM interpreter, opcodes, stack, memory	∼8,000
`core/vm/runtime/`	Standalone execution helpers	∼500
`core/state/`	StateDB, stateObject, journal	∼5,000
`core/types/`	Transaction, Block, Receipt, Log	∼4,000
`ethdb/`	Database interface (LevelDB, memorydb)	∼1,000
`params/`	Chain config, protocol params	∼2,000
`crypto/`	Keccak hashing, secp256k1 signing	∼1,500
`common/`	Address, Hash, Big types	∼2,000
`core/tracing/`	Tracing hooks interface	∼500
Subtotal (required)		**∼24,500**

TABLE II
GO-ETHEREUM MODULES NOT REQUIRED

Module	Purpose	LOC	Exclusion Reason
`p2p/`	Peer networking	∼15,000	Custom L2 networking
`eth/`	Protocol handler	∼20,000	Full node protocol
`les/`	Light client	∼10,000	Not applicable
`consensus/`	PoW/PoS consensus	∼8,000	L2 uses sequencer
`miner/`	Block mining	∼3,000	Not applicable
`cmd/`	CLI tools	∼10,000	Custom node binary
`internal/`	Internal APIs	∼5,000	Custom APIs
Subtotal (excluded)		**∼71,000**	

The estimated fork ratio is approximately 24,500 / 95,500 $\approx$ 25% of core Go code. In practice, the preferred approach imports go-ethereum as a Go module dependency rather than forking the repository, using only the required packages (Table I).

D. Fork Strategy Decision Framework

Two strategies exist for integrating Geth's EVM:

Strategy A (Import as Go module):

```
import (
  "github.com/ethereum/go-ethereum/core/vm"
  "github.com/ethereum/go-ethereum/core/state"
)
```

This provides automatic upstream updates, zero maintenance burden, and clean dependency management, but limits modification to public APIs.

Strategy B (Repository fork): Full control over all internals, including the ability to modify StateDB, add custom opcodes, and alter execution semantics. The cost is merge conflicts on upstream updates and ongoing maintenance overhead.

Based on the analysis in this paper, we recommend Strategy A. The `vm.StateDB` interface is sufficiently abstract to accommodate a custom Poseidon SMT implementation, and the `core/tracing` hooks API provides the trace capture capabilities needed for ZK witness generation.

IV. METHODOLOGY

A. Research Design

This study employs a three-phase methodology:

1) **Comparative architecture analysis**: Systematic examination of five production L2 systems to extract common patterns and identify the minimal EVM fork surface.
2) **Opcode constraint mapping**: Classification of all 75 explicitly analyzed Cancun EVM opcodes (plus ∼65 PUSH/DUP/SWAP variants, totaling ∼140) by ZK proving difficulty, estimated constraint count (R1CS and PLONKish), and recommended circuit treatment.
3) **Trace format design and benchmark projection**: Design of a selective ZK tracer using Geth's `core/tracing` hooks, with throughput projections derived from published benchmarks.

B. Opcode Constraint Classification

Each EVM opcode is classified along three dimensions:

ZK Difficulty (six tiers):

- TRIVIAL (<10 constraints): Pure stack manipulation (POP, PUSH, DUP, SWAP, PC).
- CHEAP (10–100 constraints): Simple arithmetic (ADD, SUB, MUL), comparisons, memory loads.
- MODERATE (100–1,000 constraints): Division, modular arithmetic, memory copies, log emissions.
- EXPENSIVE (1,000–10,000 constraints): Storage access (SLOAD, SSTORE), external code queries, balance lookups.
- VERY EXPENSIVE (10,000–100,000 constraints): Cross-contract calls (CALL, DELEGATECALL, STATICCALL), contract creation (CREATE, CREATE2).
- PROHIBITIVE (>100,000 constraints): KECCAK256.

Circuit Treatment (six strategies):

- DIRECT: Prove directly in arithmetic circuit.
- LOOKUP: Use precomputed lookup tables.
- ORACLE: Use preimage oracle for hash precomputation.
- SPECIAL: Dedicated sub-circuit required.
- REPLACE: Substitute with ZK-friendly alternative.

- UNSUPPORTED: Excluded from initial version.

Constraint Count: Estimated R1CS and PLONKish constraints per invocation, sourced from Polygon zkEVM ZK counter documentation [14], Scroll circuit specifications [5], and the constraint-level analysis of Burchardt et al. [3].

C. ZK Tracer Design

The tracer implements Geth's `tracing.Hooks` struct with selective callbacks:

Listing 1. ZK Tracer hook registration

```go
func (t *ZKTracer) Hooks() *tracing.Hooks {
  return &tracing.Hooks{
    OnOpcode:        t.onOpcode,
    OnStorageChange: t.onStorageChange,
    OnBalanceChange: t.onBalanceChange,
    OnNonceChange:   t.onNonceChange,
    OnLog:           t.onLog,
  }
}
```

In selective mode (default), the `OnOpcode` callback increments a counter without recording individual instructions. In full-capture mode (debugging only), it records the program counter, opcode name, remaining gas, and call depth for each instruction.

The trace output structure captures:

- **Storage accesses**: Address, slot, value for each SLOAD/SSTORE.
- **Balance changes**: Address, previous/new balance, change reason.
- **Nonce changes**: Address, previous/new nonce.
- **Log emissions**: Address, topics, data.
- **Opcode count**: Total instructions executed (always captured).
- **Contract creation**: Addresses of newly deployed contracts.

D. Benchmark Projection Methodology

Due to the Go runtime not being available on the experimental platform at the time of this study, throughput benchmarks are projected from published data using the following methodology:

1) **Base throughput**: Derived from Geth's measured live sync rate of 100–200 MGas/s [9] and standard gas costs (21,000 gas for simple transfers, 20,000–22,000 gas for cold SSTORE operations).
2) **In-memory adjustment**: A 1.5–2× multiplier accounts for in-memory StateDB eliminating disk I/O latency.
3) **Tracing overhead**: Estimated from the difference between Geth's structLogger overhead (10–50×) and the selective nature of our ZK tracer, yielding 10–30% overhead for state-change-only capture.
4) **Validation**: All projections are cross-referenced against independent benchmarks (Reth at 700 MGas/s single-threaded, Gravity Reth at 1.5 GGas/s, op-geth community benchmarks at 3,000–5,000 tx/s).

Confidence levels are assigned to each projection: HIGH (supported by multiple independent sources), MEDIUM (extrapolated from related benchmarks), or LOW (estimated from first principles with limited validation data).

V. EXPERIMENTAL SETUP

A. Implementation

The experimental implementation consists of three Go source files targeting go-ethereum v1.14.12:

- `main.go`: Benchmark suite implementing simple transfer and storage write benchmarks with configurable tracing modes (no tracing, selective ZK tracing, full opcode capture).
- `opcode_analysis.go`: Complete mapping of 75 Cancun EVM opcodes (plus PUSH1–32, DUP1–16, SWAP1–16) to ZK constraint estimates and circuit treatment recommendations.
- `setup.sh`: Build and execution script with dependency analysis.

B. Chain Configuration

The benchmark uses the Basis Network L2 chain configuration:

Listing 2. Basis L2 chain configuration

```
ChainConfig{
  ChainID:      43199,  // Basis Network
  CancunTime:   0,      // Cancun from genesis
  // No PragueTime (Avalanche constraint)
}
```

The EVM target is Cancun, consistent with the Avalanche Subnet-EVM constraint that prevents use of Pectra-era opcodes. Gas price is set to zero, matching the zero-fee enterprise model.

C. Benchmark Configurations

Five benchmark configurations are evaluated:

1) **Simple transfer, no tracing**: Baseline ETH transfers (21,000 gas) between 100 pre-funded accounts. 10,000 iterations with 10% warm-up.
2) **Simple transfer, ZK tracing**: Same workload with selective ZK tracer (state changes only, no opcode log).
3) **Storage write, no tracing**: SSTORE operations via minimal contract (8 bytes bytecode) with varying slot addresses.
4) **Storage write, ZK tracing**: Same workload with selective ZK tracer capturing StorageChange events.
5) **Full opcode trace**: Storage write workload with complete opcode capture (every instruction logged).

D. StateDB Configuration

All benchmarks use an in-memory StateDB (`rawdb.NewMemoryDatabase()`) with 100 accounts pre-funded with 1,000 ETH each. This eliminates disk I/O as a confounding variable and represents the expected deployment configuration where the L2 node maintains state in memory with periodic persistence.

E. Metrics

- **Transactions per second** (tx/s): Primary throughput metric.
- **Average transaction time** (μs/tx): Per-transaction latency.
- **Trace size** (bytes/tx): Serialized JSON trace per transaction.
- **Tracing overhead** (%): Throughput reduction attributable to tracing.
- **Memory usage** (MB): Heap allocation delta during benchmark.

VI. Results

A. Production zkEVM Architecture Comparison

Table III summarizes the architectural choices of five production systems.

A universal pattern emerges: no production zkEVM uses Geth's EVM directly for proof generation. All systems separate the execution layer from the proving layer. The execution engine (Geth, Erigon, or custom) processes transactions and generates traces; a separate prover (C++ executor, zkASM interpreter, or custom circuit) consumes these traces for proof generation.

B. EVM Execution Performance Benchmarks (Published)

Table IV presents published execution performance data used as the basis for throughput projections.

C. Opcode ZK Constraint Mapping

Table V presents the constraint costs for representative opcodes across each difficulty tier. The complete mapping covers 75 explicitly analyzed opcodes.

D. Polygon zkEVM ZK Counter Costs

Table VI presents per-opcode ZK resource consumption from the Polygon zkEVM ROM specification [14].

E. Opcode Distribution by Difficulty and Treatment

Table VII shows the distribution of the 75 explicitly analyzed opcodes across difficulty tiers and circuit treatments.

F. Projected Throughput Benchmarks

Table IX presents the projected throughput for each benchmark configuration.

All configurations with selective ZK tracing exceed the 1,000 tx/s hypothesis threshold. Even the lower bound of storage writes with ZK tracing (1,500 tx/s) provides 50% headroom above the target.

G. Geth Module Analysis

The required modules (Table I) total approximately 24,500 lines of code, representing $\sim$25% of Geth's core Go code and $\sim$5% of the total codebase ($\sim$500,000+ LOC). The excluded modules (Table II) account for $\sim$71,000 LOC of unnecessary networking, consensus, mining, and CLI infrastructure.

The op-geth reference point is instructive: Optimism achieved a complete L2 execution layer with 16,881 lines added and 664 deleted, of which only 34 lines touch the EVM itself (precompile and caller overrides). The Basis Network use case (execution + tracing, no L2 protocol features such as deposit transactions or L1 cost computation) is expected to require a smaller modification surface.

VII. Discussion

A. Key Findings

Finding 1: Execution and proving must be architecturally separated. All five surveyed production systems maintain a clear boundary between the execution engine (which processes transactions and generates traces) and the proving engine (which consumes traces and generates ZK proofs). This separation is not merely an implementation choice but an architectural necessity: the EVM's 256-bit arithmetic, stack-based execution model, and Keccak-centric hashing are fundamentally mismatched with the finite field arithmetic and polynomial constraints of ZK proof systems. Attempting to prove execution inline—without a trace intermediary—would require encoding the full EVM interpreter in a constraint system, an approach no production system has found practical.

Finding 2: Import-as-module (Strategy A) is viable for Basis Network. Optimism's op-geth demonstrates that a complete L2 execution layer can be built with only 34 lines of EVM modification. The `vm.StateDB` interface provides sufficient abstraction to integrate a custom Poseidon-based sparse Merkle tree without modifying Geth's internal state management. Scroll's experience with zktrie (and subsequent migration away from it in the Euclid upgrade) validates that tight coupling between the state trie and the EVM fork creates maintenance burden that outweighs the benefits.

Finding 3: Geth's tracing hooks API enables efficient ZK trace capture. The `core/tracing.Hooks` interface, available since Geth 1.14+, provides event-driven callbacks for all state-modifying operations. By capturing only state changes (10–30% overhead) rather than every opcode execution (10–50$\times$ overhead with structLogger), the ZK tracer achieves throughput compatible with enterprise requirements.

Finding 4: KECCAK256 is the dominant ZK constraint cost. At $\sim$150,000 R1CS constraints per invocation (approximately 1,000$\times$ the cost of Poseidon), KECCAK256 represents the single largest proving cost in any zkEVM. Two mitigation strategies are available: (a) replacing Keccak with Poseidon where possible (state trie, address derivation) and (b) using a preimage oracle with lookup tables for Keccak values that must be verified (contract storage layout, ABI encoding).

TABLE III
PRODUCTION ZKEVM AND L2 EXECUTION ENGINE ARCHITECTURES

System	EVM Approach	Fork Base	Type	Proving Time/Batch	L2 TPS
Polygon zkEVM	Custom executor (C++) + zkASM	Not Geth (custom)	Type 3	∼190–200 s	∼30–50
Polygon CDK	Erigon fork (Go)	Erigon	Type 3	∼190–200 s	∼30–50
Scroll	Geth fork (scroll-geth)	go-ethereum	Type 2	Target <30 s (2025)	Target 10K+
zkSync Era	Custom VM (EraVM)	Not Geth	Type 4	Variable	∼100–300
Optimism (op-geth)	Geth fork (minimal diff)	go-ethereum	Optimistic	N/A (no ZK)	∼2,000

TABLE IV
PUBLISHED EVM EXECUTION PERFORMANCE

Implementation	Metric	Value
Reth (live sync)	Throughput	100–200 MGas/s
Reth (single-thread)	Throughput	∼700 MGas/s
Gravity Reth (ERC20)	Transfer rate	∼41,000 tx/s
Gravity Reth (Uniswap)	Throughput	1.9 GGas/s
revmc (JIT)	Improvement	1.85–19×
Geth (standard)	Transfers	∼3,000–5,000 tx/s
op-geth (fork diff)	Lines changed	+16,881 / −664

TABLE V
REPRESENTATIVE OPCODE ZK CONSTRAINT COSTS

Opcode	Category	R1CS	PLONK	Treatment
KECCAK256	crypto	150,000	50,000	Oracle
CREATE2	system	60,000	6,000	Special
CREATE	system	50,000	5,000	Special
CALL	system	20,000	2,000	Special
DELEGATECALL	system	20,000	2,000	Special
STATICCALL	system	15,000	1,500	Special
EXP	arithmetic	5,000	500	Special
BLOCKHASH	block	5,000	500	Oracle
SELFDESTRUCT	system	5,000	500	Special
SLOAD	storage	2,000	255	Special
SSTORE	storage	2,000	255	Special
EXTCODECOPY	environment	2,000	510	Special
ADD	arithmetic	30	3	Direct
MLOAD	memory	20	2	Direct
PUSH0	stack	1	1	Direct

Finding 5: Scroll is moving away from custom zktrie. The 2025 Euclid upgrade migrates from the Poseidon-based zktrie back to standard MPT combined with OpenVM (a general-purpose zkVM). This suggests that the approach of modifying the state trie within the Geth fork—rather than abstracting it behind an interface—creates unsustainable coupling. Our Strategy A approach (custom StateDB behind `vm.StateDB` interface) avoids this pitfall.

TABLE VI
POLYGON ZKEVM PER-OPCODE ZK COUNTER COSTS

Opcode	Arith	Binary	Keccak-f	Poseidon
KECCAK256	192	193	2	10
SLOAD	0	0	0	255
SSTORE	0	–	0	255
EXTCODESIZE	0	0	0	255
EXTCODECOPY	0	–	0	510
BLOCKHASH	0	0	1	9
LOG0–LOG4	0	–	0	0

TABLE VII
OPCODE DISTRIBUTION BY ZK DIFFICULTY

Difficulty	Count	Constraints
Prohibitive	1	>100,000
Very Expensive	7	10,000–100,000
Expensive	9	1,000–10,000
Moderate	17	100–1,000
Cheap	18	10–100
Trivial	23	<10
Total (explicit)	**75**	
+ PUSH/DUP/SWAP	∼65	All Trivial
Total (all)	**∼140**	

TABLE VIII
OPCODE DISTRIBUTION BY CIRCUIT TREATMENT

Treatment	Count
Direct (arithmetic circuit)	39
Lookup (precomputed tables)	8
Oracle (preimage oracle)	2
Special (dedicated sub-circuit)	16
Replace (ZK-friendly alternative)	0
Unsupported (initial exclusion)	0

B. Invariants Discovered

The following invariants are established by this investigation and must be preserved by downstream pipeline stages:

1) **INV-EVM-1 (Execution-Proving Separation)**: The EVM executor SHALL produce execution traces consumed asynchronously by the ZK prover. No inline proving within the execution loop.
2) **INV-EVM-2 (Trace Completeness)**: Every state-modifying operation (SLOAD, SSTORE, balance change, nonce change, LOG, CREATE) SHALL appear in the execution trace. Omitting any operation invalidates the ZK witness.
3) **INV-EVM-3 (Cancun Compatibility)**: The execution engine SHALL support all Cancun EVM opcodes. No Pectra-era opcodes (PragueTime must remain unset) due to the Avalanche Subnet-EVM constraint.
4) **INV-EVM-4 (KECCAK256 Mitigation)**: KECCAK256 operations SHALL be handled via preimage oracle with lookup tables in the ZK circuit, not proved directly in R1CS or PLONKish constraints.

TABLE IX
PROJECTED EVM EXECUTION THROUGHPUT WITH ZK TRACING

Benchmark	Tracing	Projected tx/s	Trace (B/tx)	Overhead (%)	Confidence
Simple transfer	None	5,000–15,000	–	–	High
Simple transfer	Selective ZK	4,000–12,000	200–500	10–25	Medium
Storage write	None	2,000–8,000	–	–	Medium
Storage write	Selective ZK	1,500–6,000	500–1,500	15–30	Medium
Storage write	Full opcode	500–3,000	2,000–10,000	40–70	Low

5) **INV-EVM-5 (Module Import Strategy)**: The execution engine SHALL import go-ethereum as a Go module dependency (Strategy A) unless the `vm.StateDB` interface proves insufficient for Poseidon SMT integration, in which case a minimal fork (Strategy B) is acceptable.
6) **INV-EVM-6 (Selective Tracing)**: The ZK tracer SHALL capture state-modifying operations only (not every opcode) in production mode. Full opcode capture SHALL be available as a debug mode only.

C. Precompile Considerations

EVM precompiled contracts (addresses 0x01–0x0a) require special handling in the ZK circuit:

TABLE X
PRECOMPILE ZK CONSTRAINT ESTIMATES

Address	Name	Constraints	Notes
0x01	ecRecover	∼30,000	secp256k1 recovery
0x02	SHA256	∼30,000	Boolean circuit
0x03	RIPEMD160	∼30,000	Boolean circuit
0x04	identity	Trivial	Data copy
0x05	modexp	Variable	Dynamic cost
0x06	ecAdd (BN254)	∼1,000	Native to ZK field
0x07	ecMul (BN254)	∼5,000	Native to ZK field
0x08	ecPairing	∼100,000	Per pair element
0x09	blake2f	∼10,000	Compression function
0x0a	KZG eval	∼50,000	Cancun (EIP-4844)

BN254 operations (ecAdd, ecMul) are natively efficient because the ZK proof system itself operates over the BN254 curve. KECCAK256-derived precompiles (ecRecover via message hash, SHA256, RIPEMD160) inherit the high constraint cost of Boolean circuit emulation.

D. Limitations

Projected benchmarks. The throughput figures in this paper are projections derived from published data, not direct measurements. While cross-validated against multiple independent sources, actual performance on Basis Network hardware may differ. The Go benchmark code is provided for execution once the Go runtime is available.

Trace format stability. The ZK trace format defined in this paper is preliminary. The downstream witness generation module (RU-L3) will specify the exact trace schema consumed by the Rust prover, which may require modifications to the tracer output structure.

StateDB interface sufficiency. The recommendation to use Strategy A (import as module) rests on the assumption that `vm.StateDB` can accommodate a Poseidon SMT. Scroll's zktrie experience validates this assumption, but edge cases (state root computation during EVM execution, snapshot/revert semantics) may require interface extensions. If extensions prove necessary, a minimal fork targeting only `core/state` becomes the fallback.

Single-threaded execution. All projections assume single-threaded EVM execution. Parallel execution (as demonstrated by Gravity Reth at 41,000 tx/s for ERC20 transfers) could improve throughput substantially but introduces complexity in trace ordering and state consistency.

VIII. CONCLUSION

This paper investigated the feasibility of using a minimal go-ethereum fork as the EVM execution engine for the Basis Network enterprise zkEVM L2. Through systematic analysis of five production L2 systems, a complete mapping of 75 Cancun EVM opcodes to ZK constraint costs, and the design of a selective ZK tracer, we establish the following conclusions:

1) The execution-proving separation pattern is universal across production zkEVM systems. Geth's role is transaction execution and trace generation; a separate prover (in our case, a Rust-based PLONK prover communicating via gRPC) consumes these traces for proof generation.
2) A minimal Geth integration via Go module import (Strategy A) is the recommended approach, requiring approximately 24,500 LOC of Geth's core packages with zero lines of EVM modification. This is validated by Optimism's op-geth achieving a complete L2 with only 34 EVM lines changed, and Scroll's subsequent migration away from tight fork coupling.
3) The `core/tracing` hooks API provides an efficient mechanism for ZK-specific trace capture, with projected overhead of 10–30% for selective tracing versus 10–50× for full opcode capture.
4) Projected throughput of 4,000–12,000 tx/s for simple transfers and 1,500–6,000 tx/s for storage writes with ZK tracing comfortably exceeds the 1,000 tx/s hypothesis threshold.
5) KECCAK256 (∼150,000 R1CS constraints) and storage operations (255 Poseidon operations per SLOAD/SSTORE) dominate ZK proving costs. KECCAK256 mitigation via preimage oracle with lookup tables is a mandatory circuit-level optimization.

6) Six architectural invariants (INV-EVM-1 through INV-EVM-6) are established as constraints for downstream pipeline stages: the Logicist (TLA+ specification), the Architect (Go/Rust implementation), and the Prover (Coq verification).

A. Recommendations

For the Basis Network enterprise zkEVM L2 implementation:

1) Import go-ethereum v1.14+ as a Go module dependency. Use `core/vm`, `core/state`, `core/types`, `core/tracing`, and supporting packages.
2) Implement a custom `StateDB` satisfying the `vm.StateDB` interface, backed by a Poseidon-based sparse Merkle tree for ZK-friendly state management.
3) Deploy the selective ZK tracer (state changes only) in production, with full opcode capture available as a debug mode.
4) Target Geth v1.14.12+ for the Cancun EVM feature set. Do not set PragueTime in the chain configuration.
5) Monitor Scroll's Euclid migration and Geth upstream releases for tracing API changes that may affect the `core/tracing.Hooks` interface.

B. Future Work

Immediate next steps in the Basis Network R&D pipeline:

- **RU-L2 (State Management)**: Design and benchmark the Poseidon SMT implementation behind the `vm.StateDB` interface.
- **RU-L3 (Witness Generation)**: Specify the exact trace format consumed by the Rust prover and validate end-to-end trace completeness.
- **Actual benchmarks**: Execute the Go benchmark suite on target hardware to validate projected throughput figures.
- **KECCAK256 circuit**: Design and benchmark the preimage oracle with lookup tables for KECCAK256 proving.

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